

- Mirtazapine for insomnia, agitation, and gastrointestinal side effects
- Many side effects are dose-dependent (i.e., they increase as dose increases, or they reemerge until tolerance redevelops)
- Many side effects are time-dependent (i.e., they start immediately upon dosing and upon each dose increase, but go away with time)
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of duloxetine

Dosing and Use

Usual Dosage Range

- 40–60 mg/day in 1–2 doses for depression
- 60 mg once daily for diabetic peripheral neuropathic pain and fibromyalgia
- 60 mg once daily for generalized anxiety disorder
- 40 mg twice daily for stress urinary incontinence

Dosage Forms

- Capsule 20 mg, 30 mg, 40 mg, 60 mg

How to Dose

- For depression, initial 40 mg/day in 2 doses; can increase to 60 mg/day in 1–2 doses if necessary; maximum dose generally 120 mg/day
- For neuropathic pain and fibromyalgia initial 30 mg once daily; increase to 60 mg once daily after 1 week; maximum dose generally 60 mg/day
- For generalized anxiety, initial 60 mg once daily; maximum dose generally 120 mg/day



Dosing Tips

- Studies have not demonstrated increased efficacy beyond 60 mg/day
- 🔍 Some patients may require up to or more than 120 mg/day, but clinical experience is quite limited with high dosing
- In relapse prevention studies in depression, a significant percentage of patients who relapsed on 60 mg/day responded and remitted when the dose was increased to 120 mg/day
- In neuropathic pain and fibromyalgia doses above 60 mg/day have been associated with increased side effects without an increase in efficacy
- Some studies suggest that both serotonin and norepinephrine reuptake blockade are present at 40–60 mg/day
- Do not chew or crush and do not sprinkle on food or mix with food, but rather always swallow whole to avoid affecting enteric coating

- Some patients may require dosing above 120 mg/day in 2 divided doses, but this should be done with caution and by experts

Overdose

- Rare fatalities have been reported; serotonin syndrome, sedation, vomiting, seizures, coma, change in blood pressure

Long-Term Use

- Blood pressure should be monitored regularly

Habit Forming

- No

How to Stop

- Taper to avoid withdrawal effects (dizziness, nausea, vomiting, headache, paresthesias, irritability)
- Many patients tolerate 50% dose reduction for 3 days, then another 50% reduction for 3 days, then discontinuation
- 🧪 If withdrawal symptoms emerge during discontinuation, raise dose to stop symptoms and then restart withdrawal much more slowly

Pharmacokinetics

- Elimination half-life approximately 12 hours

- Metabolized mainly by CYP450 2D6 and CYP450 1A2
- Inhibitor of CYP450 2D6 (probably clinically significant) and CYP450 1A2 (probably not clinically significant)
- Absorption may be delayed by up to 3 hours and clearance may be increased by one-third after an evening dose as compared to a morning dose
- Food does not affect absorption

Drug Interactions

- Can increase TCA levels; use with caution with TCAs or when switching from a TCA to duloxetine
- Can cause a fatal “serotonin syndrome” when combined with MAOIs, so do not use with MAOIs or for at least 14 days after MAOIs are stopped
- Possible increased risk of bleeding, especially when combined with anticoagulants (e.g., warfarin, NSAIDs)
- Do not start an MAOI for at least 5 half-lives (5 to 7 days for most drugs) after discontinuing duloxetine
- Inhibitors of CYP450 1A2, such as fluvoxamine, increase plasma levels of duloxetine and may require a dosage reduction of duloxetine
- Cigarette smoking induces CYP450 1A2 and may reduce plasma levels of duloxetine, but dosage modifications are not recommended for smokers

-  Inhibitors of CYP450 2D6, such as paroxetine, fluoxetine, and quinidine, may increase plasma levels of duloxetine and require a dosage reduction of duloxetine
- Via CYP450 1A2 inhibition, duloxetine could theoretically reduce clearance of theophylline and clozapine; however, studies of coadministration with theophylline did not demonstrate significant effects of duloxetine on theophylline pharmacokinetics
- Via CYP450 2D6 inhibition, duloxetine could theoretically interfere with the analgesic actions of codeine, and increase the plasma levels of some beta blockers and of atomoxetine
- Via CYP450 2D6 inhibition, duloxetine could theoretically increase concentrations of thioridazine and cause dangerous cardiac arrhythmias

Other Warnings/Precautions

- Use with caution in patients with history of seizures
- Use with caution in patients with bipolar disorder unless treated with concomitant mood-stabilizing agent
- Rare reports of hepatotoxicity; although causality has not been established, duloxetine should be discontinued in patients who develop jaundice or other evidence of significant liver dysfunction
- When treating children, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of

nontreatment with antidepressants and make sure to document this in the patient's chart

- Distribute the brochures provided by the FDA and the drug companies
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents
- Duloxetine may increase blood pressure, so blood pressure should be monitored during treatment

Do Not Use

- If patient has uncontrolled angle-closure glaucoma
- If patient has substantial alcohol use
- If patient is taking an MAOI
- If patient is taking thioridazine
- If there is a proven allergy to duloxetine

Special Populations

Renal Impairment

- Dose adjustment generally not necessary for mild to moderate impairment
- Not recommended for use in patients with end-stage renal disease (requiring dialysis) or severe renal impairment

Hepatic Impairment

- Not to be administered to patients with any hepatic insufficiency
- Not recommended for use in patients with substantial alcohol use
- Increased risk of elevation of serum transaminase levels

Cardiac Impairment

- Drug should be used with caution
- Duloxetine may raise blood pressure

Elderly

- Some patients may tolerate lower doses better
- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older



Children and Adolescents

- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart

- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients
- Not studied, but can be used by experts

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Not generally recommended for use during pregnancy, especially during first trimester
- Nonetheless, continuous treatment during pregnancy may be necessary and has not been proven to be harmful to the fetus
- Must weigh the risk of treatment (first trimester fetal development, third trimester newborn delivery) to the child against the risk of no treatment (recurrence of depression, maternal health, infant bonding) to the mother and child

- For many patients this may mean continuing treatment during pregnancy
- Neonates exposed to SSRIs or SNRIs late in the third trimester have developed complications requiring prolonged hospitalization, respiratory support, and tube feeding; reported symptoms are consistent with either a direct toxic effect of SSRIs and SNRIs or, possibly, a drug discontinuation syndrome, and include respiratory distress, cyanosis, apnea, seizures, temperature instability, feeding difficulty, vomiting, hypoglycemia, hypotonia, hypertonia, hyperreflexia, tremor, jitteriness, irritability, and constant crying
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/atypical/antipsychotic>

Breast Feeding

- Some drug is found in mother's breast milk
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother

- For many patients, this may mean continuing treatment during breast feeding

The Art of Psychopharmacology

Potential Advantages

- Patients with physical symptoms of depression
- Patients with retarded depression
- Patients with atypical depression
- Patients with comorbid anxiety
- Patients with depression may have higher remission rates on SNRIs than on SSRIs
- Depressed patients with somatic symptoms, fatigue, and pain
- Patients who do not respond or do not remit on treatment with SSRIs

Potential Disadvantages

- Patients with urologic disorders, prostate disorders (e.g., older men)
- Patients sensitive to nausea

Primary Target Symptoms

- Depressed mood
- Energy, motivation, and interest

- Sleep disturbance
- Anxiety
- Physical symptoms
- Pain

Pearls

-  Duloxetine has well-documented efficacy for the painful physical symptoms of depression
- Duloxetine has only somewhat greater potency for serotonin reuptake blockade than for norepinephrine reuptake blockade, but this is of unclear clinical significance as a differentiator from other SNRIs
- No head-to-head studies, but may have less hypertension than venlafaxine XR
- Powerful pro-noradrenergic actions may occur at doses greater than 60 mg/day
- Not well studied in ADHD, but may be effective
-  Approved in many countries for stress urinary incontinence
- Patients may have higher remission rate for depression on SNRIs than on SSRIs
- Add or switch to or from pro-noradrenergic agents (e.g., atomoxetine, reboxetine, other SNRIs, mirtazapine, maprotiline, nortriptyline, desipramine, bupropion) with caution

- Add or switch to or from CYP450 2D6 substrates with caution (e.g., atomoxetine, maprotiline, nortriptyline, desipramine)
- Mechanism of action as SNRI suggests it may be effective in some patients who fail to respond to SSRIs

Suggested Reading

Arnold LM , Pritchett YL , D'Souza DN , et al. Duloxetine for the treatment of fibromyalgia in women: pooled results from two randomized, placebo-controlled trials . *J Womens Health (Larchmt)* 2007 ;**16** (8):1145 – 56.

Bymaster FP , Dreshfield-Ahmad LJ , Threlkeld PG , et al. Comparative affinity of duloxetine and venlafaxine for serotonin and norepinephrine transporters in vitro and in vivo, human serotonin receptor subtypes, and other neuronal receptors . *Neuropsychopharmacology* 2001 ;**25** (6):871–80 .

Hartford J , Kornstein S , Liebowitz M , et al. Duloxetine as an SNRI treatment for generalized anxiety disorder: results from placebo and active-controlled trial . *Int Clin Psychopharmacol* 2007 ;**22** (3):167–74 .

Muller N , Schennach R , Riedel M , Moller HJ . Duloxetine in the treatment of major psychiatric and neuropathic disorders . *Expert Rev Neurother* 2008 ;**8** (4):527–36 .

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Zinner NR . Duloxetine: a serotonin-noradrenaline re-uptake inhibitor for the treatment of stress urinary incontinence . *Expert Opin Investig Drugs* 2003 ;**12** (9):1559 –66.

Escitalopram

Therapeutics

Brands

- Lexapro
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: serotonin reuptake inhibitor (S-RI)
- SSRI (selective serotonin reuptake inhibitor); often classified as an antidepressant, but it is not just an antidepressant

Commonly Prescribed for

(bold for FDA approved)

- **Major depressive disorder (ages 12 and older)**
- **Generalized anxiety disorder (GAD)**
- Panic disorder
- Obsessive-compulsive disorder (OCD)

- Posttraumatic stress disorder (PTSD)
- Social anxiety disorder (social phobia)
- Premenstrual dysphoric disorder (PMDD)

How the Drug Works

- Boosts neurotransmitter serotonin
- Blocks serotonin reuptake pump (serotonin transporter)
- Desensitizes serotonin receptors, especially serotonin 1A autoreceptors
- Presumably increases serotonergic neurotransmission

How Long Until It Works

- Onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks
- If it is not working within 6–8 weeks, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses

- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission) or significantly reduced (e.g., OCD, PTSD)
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in anxiety disorders may also need to be indefinite

If It Doesn't Work

- Many patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating in depression)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)

- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer

Best Augmenting Combos for Partial Response or Treatment Resistance

- Trazodone, especially for insomnia
- Bupropion, mirtazapine, reboxetine, or atomoxetine (use combinations of antidepressants with caution as this may activate bipolar disorder and suicidal ideation)
- Modafinil, especially for fatigue, sleepiness, and lack of concentration
- Mood stabilizers or atypical antipsychotics for bipolar depression, psychotic depression, treatment-resistant depression, or treatment-resistant anxiety disorders
- Benzodiazepines
- If all else fails for anxiety disorders, consider gabapentin or tiagabine
- Hypnotics for insomnia
- Classically, lithium, buspirone, or thyroid hormone

Tests

- None for healthy individuals

Side effects

How Drug Causes Side Effects

- Theoretically due to increases in serotonin concentrations at serotonin receptors in parts of the brain and body other than those that cause therapeutic actions (e.g., unwanted actions of serotonin in sleep centers causing insomnia, unwanted actions of serotonin in the gut causing diarrhea, etc.)
 - Increasing serotonin can cause diminished dopamine release and might contribute to emotional flattening, cognitive slowing, and apathy in some patients
 - Most side effects are immediate but often go away with time, in contrast to most therapeutic effects which are delayed and are enhanced over time
- * As escitalopram has no known important secondary pharmacologic properties, its side effects are presumably all mediated by its serotonin reuptake blockade

Notable Side Effects

- Sexual dysfunction (men: delayed ejaculation, erectile dysfunction; men and women: decreased sexual desire, anorgasmia)
- Gastrointestinal (decreased appetite, nausea, diarrhea, constipation, dry mouth)
- Mostly central nervous system (insomnia but also sedation, agitation, tremors, headache, dizziness)

- Note: patients with diagnosed or undiagnosed bipolar or psychotic disorders may be more vulnerable to CNS-activating actions of SSRIs
- Autonomic (sweating)
- Bruising and rare bleeding
- Rare hyponatremia (mostly in elderly patients and generally reversible on discontinuation of escitalopram)
- SIADH (syndrome of inappropriate antidiuretic hormone secretion)



Life-Threatening or Dangerous Side Effects

- Rare seizures
- Rare induction of mania
- Rare activation of suicidal ideation and behavior (suicidality) (short-term studies did not show an increase in the risk of suicidality with antidepressants compared to placebo beyond age 24)

Weight Gain



- Reported but not expected

Sedation



- Reported but not expected

What to Do About Side Effects

- Wait
- Wait
- Wait
- In a few weeks, switch to another agent or add other drugs

Best Augmenting Agents for Side Effects

- Often best to try another SSRI or another antidepressant monotherapy prior to resorting to augmentation strategies to treat side effects
- Trazodone or a hypnotic for insomnia
- Bupropion, sildenafil, vardenafil, or tadalafil for sexual dysfunction
- Bupropion for emotional flattening, cognitive slowing, or apathy
- Mirtazapine for insomnia, agitation, and gastrointestinal side effects
- Benzodiazepines for jitteriness and anxiety, especially at initiation of treatment and especially for anxious patients
- Many side effects are dose-dependent (i.e., they increase as dose increases, or they reemerge until tolerance redevelops)
- Many side effects are time-dependent (i.e., they start immediately upon dosing and upon each dose increase, but go away with time)
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium,

a mood stabilizer or an atypical antipsychotic, and/or
discontinuation of escitalopram

Dosing and use

Usual Dosage Range

- 10–20 mg/day

Dosage Forms

- Tablet 5 mg, 10 mg (scored), 20 mg (scored)
- Oral solution 5 mg/5 mL

How to Dose

- Initial 10 mg/day; increase to 20 mg/day if necessary; single-dose administration, morning or evening



Dosing Tips

- Given once daily, any time of day tolerated
- * 10 mg of escitalopram may be comparable in efficacy to 40 mg of citalopram with fewer side effects
- Thus, give an adequate trial of 10 mg prior to giving 20 mg
- Some patients require dosing with 30 or 40 mg

- If intolerable anxiety, insomnia, agitation, akathisia, or activation occur upon either dosing initiation or discontinuation, consider the possibility of activated bipolar disorder and switch to a mood stabilizer or an atypical antipsychotic

Overdose

- Rare fatalities have been reported; symptoms include convulsions, coma, dizziness, hypotension, insomnia, nausea, vomiting, sinus tachycardia, somnolence, and ECG changes

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Taper not usually necessary
- However, tapering to avoid potential withdrawal reactions generally prudent
- Many patients tolerate 50% dose reduction for 3 days, then another 50% reduction for 3 days, then discontinuation
- If withdrawal symptoms emerge during discontinuation, raise dose to stop symptoms and then restart withdrawal much more slowly

Pharmacokinetics

- Mean terminal half-life 27–32 hours
- Steady-state plasma concentrations achieved within 1 week
- Substrate for CYP450 2C19 and 3A4
- No significant actions on CYP450 enzymes
- Food does not affect absorption



Drug Interactions

- Tramadol increases the risk of seizures in patients taking an antidepressant
- Can cause a fatal “serotonin syndrome” when combined with MAOIs, so do not use with MAOIs or for at least 14 days after MAOIs are stopped
- Do not start an MAOI for at least 5 half-lives (5 to 7 days for most drugs) after discontinuing escitalopram
- Could theoretically cause weakness, hyperreflexia, and incoordination when combined with sumatriptan or possibly other triptans, requiring careful monitoring of patient
- Possible increased risk of bleeding, especially when combined with anticoagulants (e.g., warfarin, NSAIDs)
- NSAIDs may impair effectiveness of SSRIs
- Few known adverse drug interactions

Other Warnings/Precautions

- Use with caution in patients with history of seizures
- Use with caution in patients with bipolar disorder unless treated with concomitant mood-stabilizing agent
- When treating children, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Distribute the brochures provided by the FDA and the drug companies
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents

Do Not Use

- If patient is taking an MAOI
- If patient is taking pimozide
- If there is a proven allergy to escitalopram or citalopram

Special populations

Renal Impairment

- No dose adjustment for mild to moderate impairment
- Use cautiously in patients with severe impairment

Hepatic Impairment

- Recommended dose 10 mg/day

Cardiac Impairment

- Not systematically evaluated in patients with cardiac impairment
- Preliminary data suggest that citalopram is safe in patients with cardiac impairment, suggesting that escitalopram is also safe
- Treating depression with SSRIs in patients with acute angina or following myocardial infarction may reduce cardiac events and improve survival as well as mood

Elderly

- Recommended dose 10 mg/day
- Risk of SIADH with SSRIs is higher in the elderly
- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older



Children and Adolescents

- Approved for depression in adolescents ages 12–17